Intro2CS (Fall 2026) Project Guideline

Logistics

Rules

- Form a group of 2-4 students and explore a problem of your interest.
- The problem needs to be relevant to Intro2CS course materials, e.g., topics covered in lecture slides, homework, or office hours. Relevance is broadly defined.
- Each group submits a project proposal, a poster, and a final report; see details below.
- We will select top projects for oral presentations in Week 16 and invite all students to vote for the Best Project Award.
- Your project should be original. In particular, it should not overlap with any prior or concurrent course or research project in which your group members have been involved.
- The project should not consume excessive resources—for example, by relying on a large amount of GPUs, a high-end robot, or expensive coding-tool subscriptions. Specifically:
    - Using any campus resources publicly available to all Tsinghua students, e.g., https://easycompute.cs.tsinghua.edu.cn/, is considered reasonable. Also, we ensure that each student has access to an LLM usage budget of up to RMB 1,000 (see also https://easycompute.cs.tsinghua.edu.cn/). If you have already used part of this budget before starting your project, you can contact us to have it topped up to RMB 1,000. Please provide a screenshot showing your remaining balance, taken **after this announcement was posted and before 23:59 on Nov. 11, Wednesday**. You are also welcome to use your own models or computing resources.
    - You are required to disclose the AI models and computing resources used in your project in both your poster and final report. **20% of your final report grade will be based on resource usage.** This part will be awarded by default. However, if we find that your project uses significantly more than RMB 1,000 per person in resources, but the project quality does not justify such usage, we may deduct points at our discretion based on the quality of your work.

We encourage you to explore a technical problem you are curious about and to make an original contribution, possibly with the help of generative AI. Contributions are broadly defined, including but not limited to:

- Addressing an open problem mentioned in guest lectures;
- Extending a homework problem with an original analysis;

- Investigating a failure case in an AI model;
- Fixing an open issue in a public codebase;
- Building a small gadget that improves your daily-life workflow;
- Exploring a real-world phenomenon through computational modeling;
- ...

We value the novelty of your project, your understanding of existing solutions, and the creativity of your approach. We do **not** judge your project by whether its outcome matches your original goal. You are very welcome to report failed attempts, and we value a good summary of the lessons learned from them.

Resources

https://www.xiaoda.tsinghua.edu.cn/

https://easycompute.cs.tsinghua.edu.cn/

Deliverables

All students in the same group will receive the same project proposal/poster/presentation score (40 points in total) and will only need to submit one copy to WebLearning. If we receive multiple submissions from different students within the same group, we will grade the latest submission.

Project proposal (10 points), due on Nov. 27, Friday (Week 11)

Submit a proposal as a PDF that introduces your project idea to us

- **Project background:** What problem does your project address?
- **Literature review:** What existing solutions are there, and how does your project differ from them?
- **Method:** What approach do you propose to tackle this problem?
- **Expected results:** What are your expected results?
- **AI usage disclosure:** Disclose any use of AI in preparing your proposal, including but not limited to brainstorming, literature searches, and editing or polishing your writing. We strongly recommend keeping all records of your conversations with AI to address any possible review of disputes over AI use.

We will provide a LaTeX template that you must use to generate your proposal PDF. Please list the names and student IDs of all group members in the proposal. The proposal must be at most one page long plus the AI usage disclosure and references. Submissions that use a different LaTeX template or exceed the page limit will be desk rejected.

Project poster (20 points), due on Dec. 25, Friday (Week 15)

Submit a poster as a PDF that summarizes your project. Your poster should include the following content:

- **Project background:** What problem does your project address?
- **Literature review:** What existing solutions are there, and how does your project differ from them?
- **Method:** What approach have you taken to tackle this problem?
- **Results:** What experiments have you conducted? Do the results meet the expectations stated in your proposal? In particular, did you encounter any failures during your exploration?
- **Division of labor:** What is each group member responsible for?
- **AI usage disclosure:** Disclose any use of AI in preparing your poster. We strongly recommend keeping records of all your conversations with AI in case of any review of disputes over AI use.

We will provide a PowerPoint template that you must use to generate your poster PDF. You must keep the header and font sizes specified in the template, but you may change the size, layout, and color of individual sections. Please list the names and student IDs of all group members on the poster. Submissions that do not follow the template requirements above will be desk rejected.

Oral presentation (honorary) on Dec. 30 (in-class, Week 16)

We will sort all projects by the sum of their proposal and poster scores and invite the **top six** for in-class oral presentations. Each oral presentation will be 10 minutes plus 5 minutes Q&A from the audience. All students are expected to attend the oral presentations and vote for their favorite projects; your votes will help determine the Best Project Award this year.

Project report (10 points), due on Jan. 3, Sunday (Week 16)

Submit a final report as a PDF that summarizes your project. We expect your final report to reuse the content of your proposal & poster and to provide more comprehensive details that are not included before due to space limitations.

- **Project background:** What problem does your project address?
- **Literature review:** What existing solutions are there, and how does your project differ from them?
- **Method:** What approach have you taken to tackle this problem?
- **Results:** What experiments have you conducted? Do the results meet the expectations stated in your proposal? In particular, did you encounter any failures during your exploration?
- **Division of labor:** What is each group member responsible for?
- **AI usage disclosure:** Disclose any use of AI in the whole project. We strongly recommend keeping records of all your conversations with AI in case of any review of disputes over AI use.

- **Computing resource disclosure:** Disclose the CPU/GPU computing resources you have used for the project.

We will provide a LaTeX template that you must use to generate your final report PDF. Please list the names and student IDs of all group members in the report. The report must be at most two pages long plus the AI usage disclosure, the computing resource disclosure, and references. Submissions that use a different LaTeX template or exceed the page limit will be desk rejected.

Timeline

All deadlines are 23:59 unless otherwise specified. NO LATE DAYS.

- Nov. 11, Wednesday (Week 9): Project guideline released.
- Nov. 27, Friday (Week 11): Project proposal due.
- Dec. 25, Friday (Week 15): Project poster due.
- Dec. 28, Monday (Week 16): Oral presentations announced.
- Dec. 30, Wednesday in-class (Week 16): Oral presentations.
- Jan. 3, Sunday (Week 16): Project report due.